PHYS 580 Homework 2 (Chap.3)

All questions are worth 10 points, irrespective of complexity or length. Please submit all your solutions on Brightspace in PDF format, together with the source code (if any). Make sure to

a) discuss the physics and your results, as appropriate and/or directed,
b) include graphical output to illustrate the results,
c) include the source codes of your programs (at least the critical parts thereof),
d) describe briefly what your program does, and how it does it (algorithm), and
e) state the nature of the numerical approximation you used and demonstrate how you know that the approximation (with the parameters you used) is adequate for the problem you used it for.

Note the errata pages for the textbook: http://www.physics.purdue.edu/~hisao/book/www/errata.pdf
which are explained further on http://www.physics.purdue.edu/~hisao/book/www/errata.html

1. For the driven, nonlinear pendulum, investigate how the Poincare section depends on the strength of the linear dissipation term. Use the same physics parameters as the calculation in Fig. 3.6 of the Giordano-Nakanishi book.
   a. First set $F_D = 0.5$, $q = 0.5$, which puts you in the periodic regime, and then vary q to see how the Poincare section changes.
   b. Next, repeat the same from a starting point $F_D = 1.2$, $q = 0.5$ in the chaotic regime.

   In both cases, keep all other physics parameters fixed as you vary q.

2. Problem 3.4 (p.53).
   For simple harmonic motion, the general form for the equation of motion is

$$d^2x \,/\, dt^2 = -\,k\ \mathrm{sgn}(x)\ |x|^{\alpha}, \qquad (3.9)$$

   with $\alpha = 1$, where the sign function is: $\mathrm{sgn}(x) = \begin{cases} +1, & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1, & \text{if } x < 0 \end{cases}$.

   This has the same form as (3.2), although the variables have different names. Begin by writing a program that uses the Euler-Cromer method to solve for x as a function of time according to (3.9), with $\alpha = 1$ (for convenience you can take $k = 1$). The subroutine described in this section can be modified to accomplish this. Show that the period of the oscillations is independent of the amplitude of the motion. This is a key feature of simple harmonic motion.

   Then expand your program to treat the case $\alpha = 3$. This is an example of an *anharmonic* oscillator. Calculate the period of the oscillations for several different amplitudes (amplitudes in the range of 0.2 to 1 are good choices), and show that the period of the motion now depends on the amplitude. Give an intuitive argument to explain why the period becomes longer as the amplitude is decreased.

3. Problem 3.5 (p.54).
   For the anharmonic oscillator of (3.9) in the previous exercise, it is possible to analytically obtain the period of oscillation for general values of α in terms of certain special functions.
   a. Do this, and describe how the relationship between the period and the amplitude depends on the value of α. Can you give a physical interpretation to the finding?
   b. Also check how well the periods computed in Problem 2 agree with the analytic results and discuss whether the difference you find is within the expected accuracy of the numerical calculation.

   Hint: If you multiply both sides of (3.9) by dx/dt, you can integrate with respect to t. This then leads to a relation between the velocity and x.

4. Problem 3.13 (p.65).
   Write a program to calculate and compare the behavior of two, nearly identical pendulums. Use it to calculate the divergence of two nearby trajectories in the chaotic regime, as in Figure 3.7, and make a qualitative estimate of the corresponding Lyapunov exponent from the slope of a plot of $\log(\Delta\theta)$ as a function of t.

5. Problem 3.20 (p.70).
   Calculate the bifurcation diagram for the pendulum in the vicinity of $F_D = 1.35$ to 1.5. Make a magnified plot of the diagram (as compared to Figure 3.11) and obtain an estimate of the Feigenbaum δ parameter.